

Continuous In-Line Virus Inactivation for Next Generation Bioprocessing

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Viral inactivation plays a critical role in assuring the safety of monoclonal antibody (mAb) therapeutics. Traditional viral inactivation involves large holding tanks in which product is maintained at a target low pH for a defined hold time, typically 30–60 min. The drive toward continuous processing and improved facility utilization has provided motivation for development of a continuous viral inactivation process. To this end, a lab-scale prototype viral inactivation system was designed, built, and characterized. Multiple incubation chamber designs are evaluated to identify the optimal design that enables narrow residence time distributions in continuous flow systems. Extensive analysis is conducted supporting rapid low pH viral inactivation and included evaluations with multiple viruses, a range of pH levels, buffer compositions, mAb concentrations, and temperatures. Multiple test conditions are evaluated using the in-line system and results compared to traditional batch-mode viral inactivation. Comparability in kinetics of virus inactivation suggests equivalency between the two approaches.

1. Introduction

Traditional batch processing methods for mAb production are being replaced by intensified, connected and/or fully continuous processes as well as combinations of batch and continuous in efforts to streamline production, enable flexible manufacturing, increase efficiency and productivity, and reduce facility footprint.^[1–4] Newer processing templates for mAb production eliminate intermediate hold tanks wherever possible and streamline steps such that various unit operations can be connected in series with minimal operator handling.

In a typical mAb production process, a viral inactivation step with low pH is implemented between the Protein A capture and the subsequent chromatographic polishing steps. The purpose of this step is to inactivate any enveloped viruses (e.g., retrovirus) that may be present, and it provides a significant orthogonal contribution to viral safety assurance in biological products.^[5–7] The robustness of the low pH inactivation step is well-established and standardized for batch mode operation.^[8–11]

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In traditional low pH viral inactivation, the product eluted from the primary capture purification step is transferred into a large holding tank where the pH is adjusted to a defined target, typically between pH 3.5 and 4.0, with mixing. This material is then transferred to a second large holding tank where the low pH is maintained for 30–60 min. The purpose of the transfer is to reduce the risk associated with any droplets on the wall of the first tank that may not have been exposed to the low pH conditions.^[12] As a result, this step requires a large facility footprint from the two holding tanks, and a considerable amount of operator handling and long process times. Furthermore, sampling and sample processing time can also be significant.

The batch nature of this step presents challenges for continuous processing and recent efforts have focused on modifying traditional operations to better integrate

low pH viral inactivation into continuous processing manufacturing.^[13–15]

Although a 30–60 min hold is the current industry standard for low pH incubation,^[9] there is evidence demonstrating that viral inactivation occurs within a much shorter time.^[12] Studies have shown that Xenotropic Murine Leukemia Virus (XMuLV) is rapidly inactivated at pH 3.6 and that inactivation is generally insensitive to a range of parameters, including mAb concentration, buffer type, and acid titrant.^[16] A challenge in implementing rapid viral inactivation in a continuous flow process is the design of an incubation chamber that provides sufficient target contact time.^[17] Incubation chamber design is critical for overcoming the axial dispersion that occurs when concentrated protein solutions move through tubing.^[18] Assuming ideal plug flow conditions, a pulse of fluid would be expected to exit the system at the same concentration and velocity as at the inlet, but axial dispersion occurs along the chamber length where the initial pulse is subjected to a differential velocity profile that becomes parabolic in shape as flow develops along the length of the tubing (Figure 1A), also referred to as Poiseuille flow. The implication of axial dispersion can be observed in the concentration profile at the tubing outlet: an initial pulse injection may be very narrow, but with varying residence times the concentration profile spreads out into a wide distribution of molecule species. With respect to patient safety, ensuring the product is exposed to a minimum residence time is paramount,

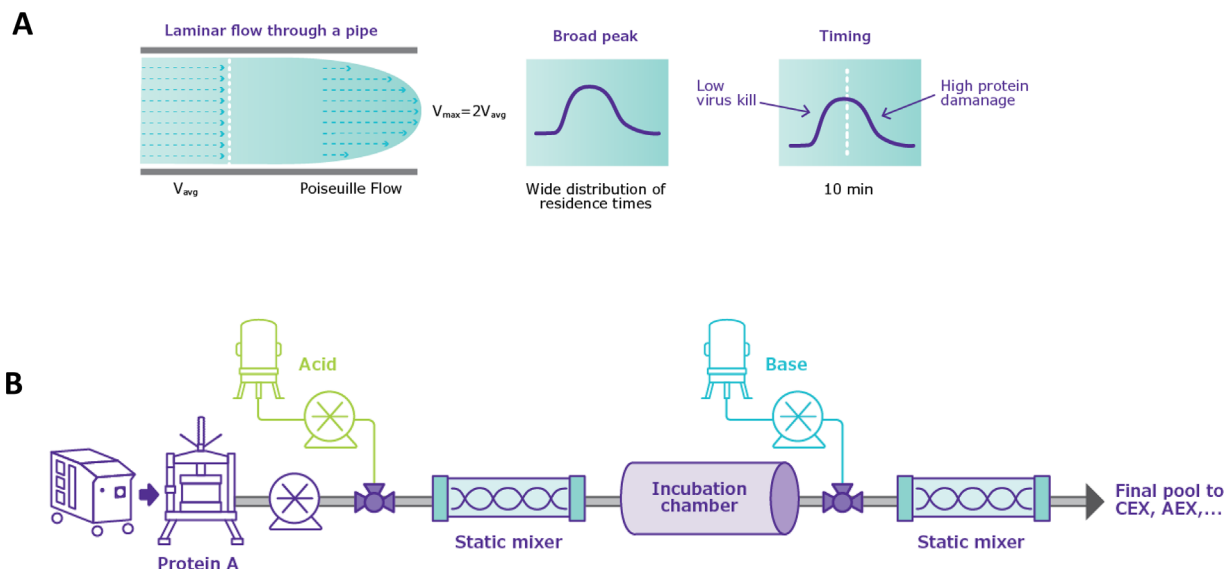


Figure 1. Cartoon representations of fluid transport in a tube and the resultant concentration at the outlet of the tube (A), Virus inactivation experimental protocol schematic (B).

where any part of the fluid that exits the chamber earlier than the prescribed time could result in a lower inactivation level than expected. To overcome this dispersion, the system residence time must be extended to ensure inactivation spanning the wide distribution of residence times, i.e., the tube must be lengthened considerably to account for the dispersion.

An ideal incubation chamber design would be a plug flow reactor in which all contents of the reactor experience the identical residence time. This is desirable for in-line viral inactivation as the virus would be exposed to the low pH conditions for a short amount of time without the need for a large safety factor to account for a spread in the residence time distribution. This more rapid step may be advantageous for mAb quality considerations as the protein would have minimal exposure to the harsh low pH conditions.

In this manuscript, a continuous in-line viral inactivation operation is described which eliminates many of the bottlenecks associated with the traditional batch viral inactivation operation. The continuous process does not require large holding tanks, thereby reducing facility footprint and takes advantage of the rapid inactivation kinetics to reduce the time required to ensure viral inactivation. Several incubation chambers which provide narrow residence time distributions approaching plug flow performance, were designed and evaluated. While safety factors and regulatory considerations will dictate future viral inactivation time requirements, it is possible that in-line viral inactivation can be much shorter than traditional viral inactivation operations and can be integrated into a continuous manufacturing process.

2. Experimental Section

2.1. Chemicals/Reagents

Stock solutions of Tris base, acetic acid, citric acid, and sodium acetate buffer were prepared using reagents from Fisher

Scientific (Hampton, NH) and MilliporeSigma (Burlington, MA). Eshmuno[®] A Protein A resin and riboflavin were obtained from MilliporeSigma and hIgG was obtained from SeraCare (Milford, MA).

Phi6 bacteriophage and host bacteria, *Pseudomonas pseudoalcaligenes* S4, were kindly provided by Dr. Clifton Dowell, Department of Microbiology University of Massachusetts, Amherst. *P. pseudoalcaligenes* S4 was prepared by inoculating Phi6 liquid media (10 g L⁻¹ Bacto Tryptone, 5 g L⁻¹ Bacto Yeast Extract, and 5 g L⁻¹ NaCl) overnight at 25 °C. Phi6 bacteriophage was propagated in *P. pseudoalcaligenes* overnight at 25 °C. After overnight incubation, cells and cell debris were pelleted by ultracentrifugation at 12 000–16 000 $\times$ g and virus particles were eluted from cell lysates using buffer (40 mM Tris base, 150 mM NaCl, and 10 mM MgCl₂·6H₂O) at room temperature for 30 min. The Phi6 virus particles were concentrated by ultracentrifugation and the pellet was resuspended in Phi6 assay buffer (21 mM dibasic sodium phosphate, 11 mM monobasic potassium phosphate, 43 mM NaCl, 9.3 mM NH₄Cl, 0.4 mM g L⁻¹ magnesium sulfate, pH 6.5). The concentrated stock was 0.22 μ m filtered (Millex-GV syringe Filter Unit). Bovine Serum Albumin (BSA) was added to the purified and concentrated Phi6 stock to a final concentration of 0.5% (v/v) and was aliquoted and stored in at –80 °C.

XMuvLV was obtained from Charles River Labs (Malvern, PA) and PRV was obtained from ATCC (ATCC[®] VR-135[™]). XMuvLV stocks were generated by infecting confluent Mv1Lu cells (ATCC[®] CCL-64[™]) in Advanced/F12 Dulbecco's Modified Eagle's Medium (DMEM) (Thermo Fisher Scientific, Waltham, MA) containing 2.5% (v/v) fetal bovine serum (FBS) with penicillin (0.2 U mL⁻¹), streptomycin (0.2 μ g mL⁻¹), 0.1 mM Non-Essential Amino Acids (NEAA), and 2 mM L-glutamine. Following incubation at 37 °C in 5% CO₂ for 3 days, the medium was replaced with the same medium without serum and infection continued for a further three days. Cell culture harvest was centrifuged at 300 $\times$ g for 20 min and supernatant was

filtered using 0.45 µm Stericup® PVDF membrane filter (MilliporeSigma). Virus was concentrated by centrifugation at 12 390 $\times$ g for 2 h at 4 °C using a SLC 6000 rotor in a Sorvall centrifuge. Virus pellets were resuspended in TNE buffer (10 mM Tris, 100 mM NaCl, 1 mM EDTA, pH 7.5) and stored at –80 °C. PRV stocks were generated by infecting confluent CRFK cells (ATCC® CCL-94™) in DMEM (Thermo Fisher Scientific) containing 2.5% (v/v) FBS with penicillin (0.2 U mL^{–1}), streptomycin (0.2 µg mL^{–1}), 0.1 mM NEAA, and 2 mM L-glutamine. Following incubation for three days, cell lysate was clarified by centrifugation at 862 $\times$ g for 20 min followed by filtration with a 0.45 µm Stericup® PVDF membrane filter. Virus was concentrated using Centricon® Plus 70 Ultracel® membrane devices (MilliporeSigma) and centrifugation at 2390 $\times$ g (Thermo Scientific Legend XTR centrifuge). Virus was further concentrated by ultracentrifugation using SW28TI rotor in a Beckman L90K centrifuge. Virus pellets were resuspended in TNE buffer and stored at –80 °C.

The mAb (IgG1 kappa) used during these studies was produced in CHO cells and harvested via two-stage depth filtration. The mAb was purified using Eshmuno® A Protein A resin and buffer exchanged into the different representative buffer conditions.

2.2. Virus Titer Assays

Inactivation studies were conducted using three viruses, bacteriophage Phi6, Xenotropic Murine Leukemia Virus (XMuLV), a murine retrovirus, and Pseudorabies Virus (PRV), a swine alphaherpesvirus closely related to human herpesviruses, all of which are enveloped viruses, known to be sensitive to low pH treatment, which likely acts by damage to viral transmembrane proteins.^[5,6,8]

Phi6 was used as a model virus for the screening studies as it offers rapid titration and has reduced biosafety requirements compared to the mammalian enveloped viruses.^[19,20] XMuLV and PRV are model enveloped viruses commonly used for clearance studies for mAb production processes.

Phi6 titers were determined using the standard plaque assay method and expressed as pfu mL^{–1}.^[21–23] For this assay, 300 µL of sample and 100 µL of host cells were mixed with tempered (50 °C) Phi6 overlay agar (7 g L^{–1} Bacto agar, 10 g L^{–1} Bacto Tryptone, 5 g L^{–1} Bacto Yeast Extract, 5 g L^{–1} NaCl) and plated onto Phi6 agar plates (10 g L^{–1} Bacto agar, 10 g L^{–1} Bacto Tryptone, 5 g L^{–1} Bacto Yeast Extract, 5 g L^{–1} NaCl) and incubated overnight at 25 °C. XMuLV and PRV titers were determined by tissue culture infectious dose 50% TCID₅₀ assays^[24] using PG-4 (ATCC® CRL-2032™) or CRFK indicator cells, respectively. Test samples were diluted 1:50 to mitigate cytotoxicity with subsequent serial dilution in cell culture media. Then, 100 µL aliquots of each dilution were added to a 96-well plate containing indicator cells. Following incubation at 37 °C in 5% CO₂ for 7 days, infected wells were visually assessed for cytopathic effect (CPE) and titers were calculated using standard methods based on the frequency of infection at particular dilutions.^[5] Log reduction values (LRVs) were determined as the difference in titer between the test material and the control.

2.3. qPCR Virus Quantitation Methods

While the infectivity assays determined the extent of virus inactivation, quantitative polymerase chain reaction (qPCR) amplification was employed to quantitate intact viral (XMuLV) particles in the sample post-processing. In this case, qPCR was used to confirm virus particles, shown to be inactivated via the infectivity assay, were indeed present in the samples.^[25] XMuLV RNA was extracted from 100 µL of sample using PureLink® Viral RNA/DNA Mini Kit (Thermo Fisher Scientific) according to the manufacturer's protocol, with some modifications.^[25] Nuclease digestion using 0.2 U µL^{–1} of micrococcal nuclease for 1 h at 37 °C was performed directly on the samples to digest free nucleic acid. After digestion, EGTA (60 mM) was added to quench the nuclease activity and 50 fg of non-relevant Phi-X174 DNA was spiked into the mixture to monitor nucleic acid recovery through the extraction process. The nucleic acids were then purified according to manufacturer's protocol and eluted from the spin columns in 100 µL of nuclease-free water.

Following extraction, qPCR reactions were performed using the ABI ViiATM7 Fast qPCR instrument and analyzed using the ViiATM7 RUO software. XMuLV Reverse Transcriptase-qPCR (RT-qPCR) comprising 1X TaqMan™ Fast 1-Step Fast Master Mix (ABI) 300 nM of forward (5'-CAGGAGCCTCGGTA-CAACGT-3') and reverse (5'-TTGGTAACTCTCCAAGTAA-CATTGA-3') primers and 250 nM of XMuLV probe (5'-6FAM-ACAGCCCTCACCAGA-MGBNQG 3').^[26] Phi-X qPCR assays were run in parallel to the XMuLV assays and utilized 1X TaqMan™ Fast Advanced Master Mix (ABI) with 200 nM of forward (5'-GCTGACAACCGTCCTTTACTTG-3') and reverse (5'-TGGTCAGTTCCATCAACATCATAGC-3') primers and 150 nM of PhiX probe (5'-6FAM-ATGCGCTCTAATCTC-MGBNFQ-3'). The total reaction volume was 10 µL and the cycling conditions consisted of an initial cDNA synthesis step of 50 °C for 5 min, denaturation step of 95 °C for 20 s and 40 cycles of 95 °C for 3 s, and 60 °C for 30 s. To determine the number of viral genome copies in the test samples, the quantification cycle (Cq) value of the test sample was compared to the Cq values of a standard curve. In-house qPCR standards were generated from in-vitro transcribed RNA using T7 polymerase from a recombinant plasmid containing XMuLV sequence. Results for XMuLV quantitation were considered acceptable if the levels of non-relevant target indicated nucleic acid recovery of at least 50%.

2.4. Batch Viral Inactivation Procedure

Initial batch studies evaluated XMuLV and PRV inactivation using mAb in 50 mM acetate buffer at high (30 g L^{–1}) and low (5 g L^{–1}) concentrations, at pH 3.5 and 3.7 at 23 ± 1 °C. Subsequent studies focused on XMuLV inactivation in ≈20 g L^{–1} mAb with different elution buffers of acetate, citrate, or glycine, each at 50 mM and at two different temperatures of 23 ± 1 °C and 17 ± 1 °C using pH 3.7. The later studies also focused on shorter incubation times as complete inactivation was observed within 5 min in the initial studies. The experimental design used for the batch studies is shown in Figure 1B. Aliquots of the mAb were prepared and labeled as Control or Test Material. Briefly, in the

batch studies, the pH of the Test Material was first titrated to the target pH using 8.7 M acetic acid for the acetate elution pools or 1 N hydrochloric acid for the citrate and glycine elution pools. The pH was adjusted to within 0.1 pH units below the target to account for a slight pH increase observed after the pH 8.0 virus spike is added. For the Control, the pH was maintained at the initial pH (pH 6.2 for Phi6, pH 5.0 for XMuLV and PRV), but were diluted similarly to Test Material using Tris buffer, pH 7.5. Virus was then spiked into the Control and Test Material to a target titer of 1×10^7 pfu mL (0.15% (v/v) Phi6 spike), or 1×10^7 TCID₅₀/mL (5% (v/v) XMuLV spike, 10% (v/v) PRV spike) and vortexed to ensure a homogeneous mixture. After each incubation timepoint, 1 mL was removed and transferred to a tube containing of 2 M Tris, pH 11, to neutralize (pH > 6) the sample and quench the inactivation step. The pH was measured using a calibrated pH meter (Thermo Fisher Oakton 110 Series) before and after the addition of virus. Each sample was immediately diluted in either Phi6 assay buffer or cell culture media depending on the virus and then assayed. For the Control, an equivalent volume of 25 mM Tris, pH 7.5, was added to the sample instead of the base solution. Neutralized samples were assayed using the plaque (Phi6) or TCID₅₀ assay (XMuLV and PRV).

2.5. Titration Curve Generation

Titration curves were generated for a variety of potential buffer and mAb systems. Titration curves were generated at room temperature, $23 \pm 1^\circ\text{C}$. A reservoir containing each buffer was titrated with 1 M acetic acid. The low pH target was $\text{pH } 3.5 \pm 0.1$. A 25 mL aliquot of each test case was titrated with 1 M acetic acid to reach a target of pH 3.4 to cover the full pH target window. A separate 25 mL aliquot was titrated to reach the target of pH 3.5 precisely and then titrated with 2 M Tris base to reach a target of pH 8. Titration curves were also generated in the presence of mAb.

2.6. Residence Time Distribution Testing for Incubation Chamber Designs

An experimental test set-up was built for evaluating residence time distribution for different incubation chamber designs. The system consisted of a peristaltic pump for the main flow through the tubing with flow rates that ranged from 1 to 10 mL min⁻¹, a sample injection loop, purge port, an interchangeable incubation chamber, and an in-line UV absorbance detector (Reach Devices RD4-250-280-RI) at the tube outlet. Consistent mixing is critical in the design of the continuous flow chamber.^[14] The in-line static mixers included a sufficient number of mixing elements at each mixing point based on manufacturer recommendations and the experimental flow properties. Preliminary tests using buffer solutions were used to test the mixing and ensure that all pH target setpoints were achieved. The system was initially filled with water followed by a marker species of saturated riboflavin solution in water or 1 mg mL⁻¹ hIgG in Tris buffer (50 mM Tris, 150 mM NaCl, pH 7.4). The sample volume ranged from 0.5 to 5 mL depending on the size of the sample loop. Water

was then pumped through the system, which pushed the marker species through the tubing and UV absorbance detector at the end of the tube. The incubation chamber consisted of 250" of polymer tubing (1/8" ID). The tubing type varied depending on the incubation chamber setup – Tygon® (Masterflex) for straight and serpentine geometries, reinforced silicone tubing (Cole-Parmer) for a mechanical method, and PTFE tubing (McMaster-Carr) for an immiscible fluid method. The signal from the UV detector, as a function of time, was used to determine the marker species' residence time in the system. The UV pulse data were integrated to calculate cumulative distribution curves as a function of volume.^[18] The resultant cumulative distribution curves were then used to better characterize and compare different chamber designs. For a thorough discussion on this type of analysis the reader is directed to the work of Orozco et al.^[14]

Different options were explored for incubation chamber design: a) the simplest option was a straight length of tubing connected to a sample port and detector; (b) the tubing was coiled to induce secondary flow formation; (c) a mechanical method was used to create and translate packets of fluid along the tubing by compressing the tubing with rollers; (d) an immiscible fluid was used to create the packets of fluid and air bubbles were injected on one or both sides of the marker species sample and were removed by a de-bubbler at the outlet of the incubation chamber but prior to the UV absorbance detector. Because of differences in sizes of incubation chambers all data was normalized to the V50 value, the volume at which 50% of the injected pulse has exited the chamber as measured by the detector.

2.7. Viral Inactivation System

A lab-scale in-line low pH viral inactivation system was constructed comprising on-line process pH sensors (S272CD) and transmitters (TX100) obtained from Sensorex. The programmable peristaltic pump (NE-9000G-UP) and syringe pumps (NE-501L) were obtained from New Era Pump Systems, Inc. Syringes (model 1100) were obtained from Hamilton Company, static mixers from StaMixCo LLC (HT-40-3.18-12-PP) and Analytical Scientific Instruments US, Inc. (421-0500B), and pressure sensors from PendoTECH. The main feed line of the system consisted of Tygon® tubing (1/8" ID). The pumps and sensors were integrated into a LabVIEW (National Instruments) program which provided control of all pump flow rates and real-time output of the UV signals, pH responses, and pressure readings. The Test Material was pumped through the system and the acid and base additions were controlled to deliver volumes determined using batch titration studies.

The system was designed for incoming feed flow rates of 1–10 mL min⁻¹. An acid addition line connected into the main feed line and was controlled by a syringe pump. After acid addition, the Test Material passed through an in-line static mixer and then through an incubation chamber where it was maintained at low pH (as monitored via an in-line pH probe both pre- and post-incubation chamber) for a target inactivation time; different lengths of tubing were used in the incubation chamber to accommodate the various incubation time targets. At

the outlet of the incubation chamber, the Test Material was contacted with the base addition line which was controlled by another syringe pump and then passed through another in-line static mixer. The Control was also passed through the in-line system with in-line buffer additions (Phi6 assay buffer or Tris buffer) instead of acid and base additions.

2.8. In-Line Viral Inactivation Procedure

In-line virus inactivation studies were carried out in Protein A elution pools at ambient temperature, $23 \pm 1^\circ\text{C}$. Neutralized samples were immediately diluted in either Phi6 assay buffer or cell culture media and assayed. The Control was similarly pumped through the in-line system. However, the pH was maintained at the initial pH values (pH 6.2 for Phi6), (pH 5.0 for XMuLV and PRV). To maintain the same dilution ratio as the Test Material, the Control was similarly diluted with buffer. The Control was critical to compare reduction values for the Test Material and assess any effects of the in-line system itself on virus titer, comparing the virus titer before being pumped through the system with the titer of the longest processing timepoint.

3. Results

3.1. Batch-Mode Viral Inactivation

The inactivation kinetics of several viruses were studied under a matrix of experimental conditions to determine the impact of different parameters including pH, mAb concentration, buffer type, and temperature.

As expected, pH has a considerable effect on virus inactivation. The effect of pH on XMuLV inactivation kinetics at ambient temperatures and in acetate buffer is summarized in **Figure 2A**. Complete XMuLV inactivation is observed within 1 min at pH 3.5 and 3.7 while inactivation kinetics are slower at the higher pH of 4.0 and 4.2. However, five logs of inactivation are still achieved within 5 min.

The impact of mAb concentration on viral inactivation kinetics was also examined. XMuLV and PRV inactivation at pH 3.5 and 3.7 were evaluated at 5 and 30 g L^{-1} mAb in acetate buffer. XMuLV inactivation is summarized in **Figure 2B** and PRV inactivation is summarized in **Figure 2C**. The effect of mAb concentration was negligible under this set of conditions for both viruses: complete XMuLV inactivation occurred within 5 min at both pH levels and mAb concentrations and for PRV, complete inactivation is observed within 0.3 min at both pH levels and mAb concentrations.

Several buffers, representative of typical Protein A elution buffers, were evaluated to assess their impact on viral inactivation. Results of XMuLV inactivation studies at pH 3.7 at $23 \pm 1^\circ\text{C}$ in 50 mM acetate, citrate, or glycine buffer is shown in **Figure 2D**. Inactivation in acetate buffer was slightly better than in glycine buffer and noticeably more robust than in citrate buffer.

Temperature is also considered an important parameter impacting the efficacy of viral inactivation. XMuLV inactivation

was assessed in either acetate, citrate, or glycine buffer at $17 \pm 1^\circ\text{C}$ and $23 \pm 1^\circ\text{C}$. The results show slightly slower inactivation within the first 15 min of incubation at the lower temperature, regardless of buffer type, **Figure 2E**. Even though good reduction values were achieved within 3 min, residual infectivity was observed at the later timepoints at the lower temperature.

The effect of buffer composition was also assessed for PRV inactivation. Since PRV has been shown to be easily inactivated, these studies were carried out at the worst-case lower temperature only ($17 \pm 1^\circ\text{C}$) to differentiate the effect of buffer. Results show buffer type has minimal impact on PRV inactivation kinetics, **Figure 2F**. LRVs greater than 5 were obtained within 1 min in each buffer.

3.2. Titration Results

To determine acid and base volumes required to achieve target pH values, static pH titrations were performed under a matrix of representative conditions to reflect a range of Protein A elution pool conditions: different mAb concentrations, buffer types, and buffer strengths. The low pH target was $\text{pH } 3.5 \pm 0.1$. The acid titration results are summarized in **Figure 3**. Antibodies inherently possess many potential buffering groups within their amino acid side chains.^[27] These side chains have pKa values in the 3.7–6.5 range and therefore can provide buffering capacity throughout the pH range often used in mAb processing. In the current study, an effect of mAb concentration on titrant volume was observed. As expected, the effect of mAb concentration was more pronounced at lower buffer concentration, as indicated by the wider range of acid volumes needed to achieve a target pH level for 50 mM acetic acid (**Figure 3A**) than for higher buffer concentrations (**Figure 3C**).

After achieving the low pH target value, the Test Material was then neutralized to pH 8.0 using 2 M Tris base (**Figure 3E**). The mAb concentration change was monitored and experienced a 40–68% overall dilution from the acid and base additions depending on the initial buffer conditions. This represents a “worst-case” dilution scenario since the initial buffer pH levels were slightly higher than typical Protein A elution pool levels and the buffer strength was stronger than typical elution pool levels due to mixing with equilibration buffer.

The static titration values were used as a guide for selecting acid and base set points to reach different target pH values in the in-line system under various conditions in the continuous VI setup. For the in-line system, the target pH after the acid addition was pH 3.5. The target pH after the base addition was pH 7.0–8.0 and the target dilution percentage due to the acid and base additions was less than 20%. The continuous virus inactivation system met pH and dilution targets as the amount of acetic acid required to achieve the target pH value differed by less than 10% between the batch and in-line systems.

3.3. Residence Time Distribution Mapping

Incubation chamber designs included straight tubing, coiled tubing, and several other designs intended to maintain narrow

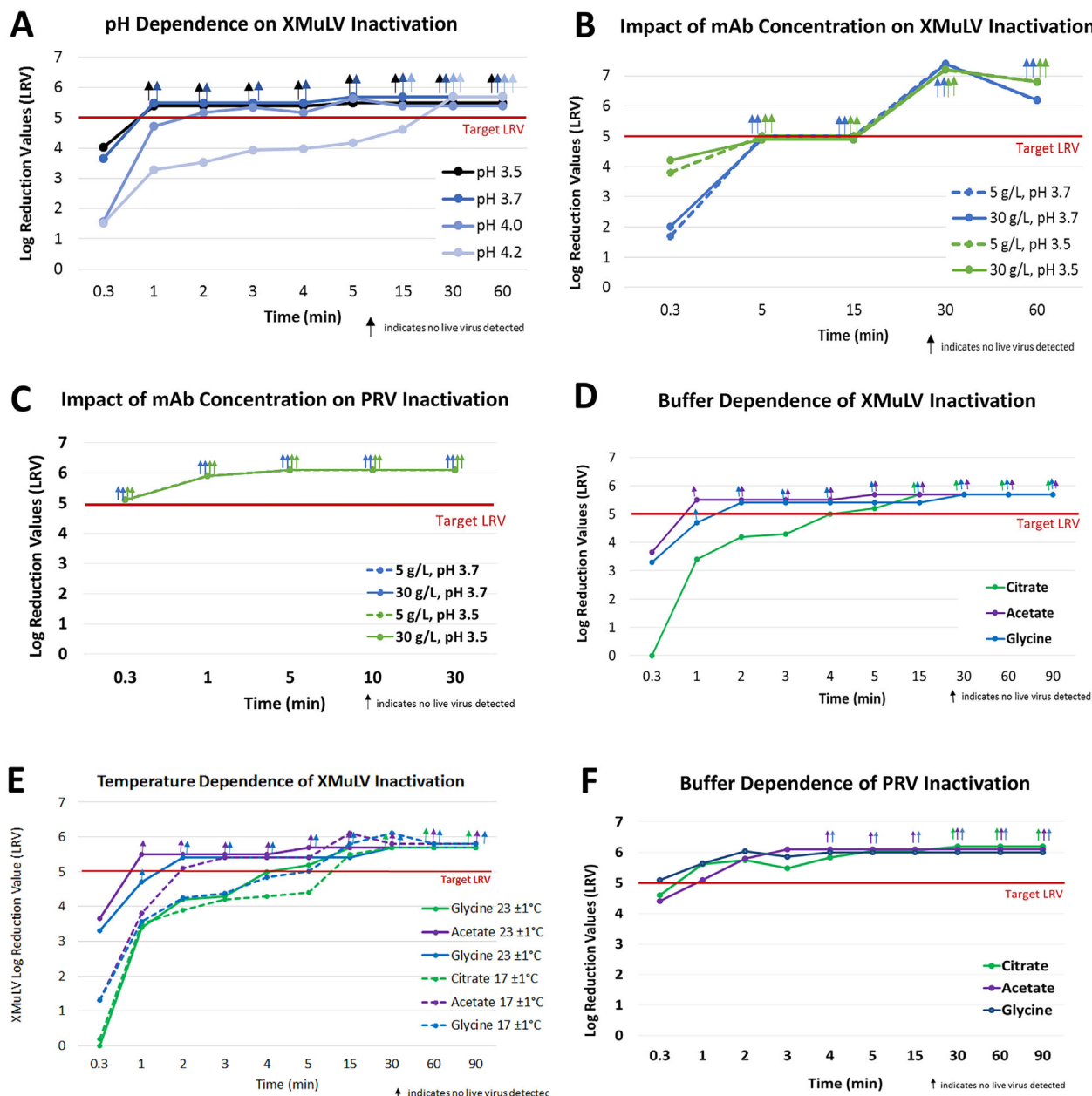


Figure 2. Batch virus inactivation results as a function of different parameters shown as log reduction values (LRVs). Arrows indicate that no live virus was detected. A) Effect of pH on XMuLV inactivation; pH 3.5–4.2, (B) Effect of mAb concentration on XMuLV inactivation; 5 vs. 30 g L⁻¹ mAb, (C) Effect of mAb concentration on PRV inactivation; 5 vs. 30 g L⁻¹ mAb, (D) Effect of buffer type on XMuLV inactivation; citrate vs. acetate, vs. glycine, pH 3.7, (E) Effect of temperature on XMuLV inactivation; 17 ± 1 °C vs. 23 ± 1 °C, pH 3.7, (F) Effect of buffer type on PRV inactivation; citrate vs. acetate vs. glycine, pH 3.7, 17 ± 1 °C.

residence time distributions. The results from pulse injections with riboflavin were plotted as cumulative distribution as a function of volume (normalized to V50, i.e., volume where 50% of the pulse injection has exited the chamber) to compare different designs, and are shown in **Figure 4**. A curve representing ideal plug flow (narrow residence time) is shown for comparison. Many of the designs offer significant improvements over the simplest straight tubing design (wide range of residence time) and approach the profile of the ideal plug flow.

3.4. In-Line Virus Inactivation

Straight tubing was used as the incubation chamber with the in-line viral inactivation system. Inactivation from the in-line system was compared to the batch inactivation results. These studies were performed at room temperature (23 ± 1 °C) in acetate buffer. Under ideal conditions (i.e., perfect mixing, identical pH conditions, etc.), the batch and in-line results should be identical.

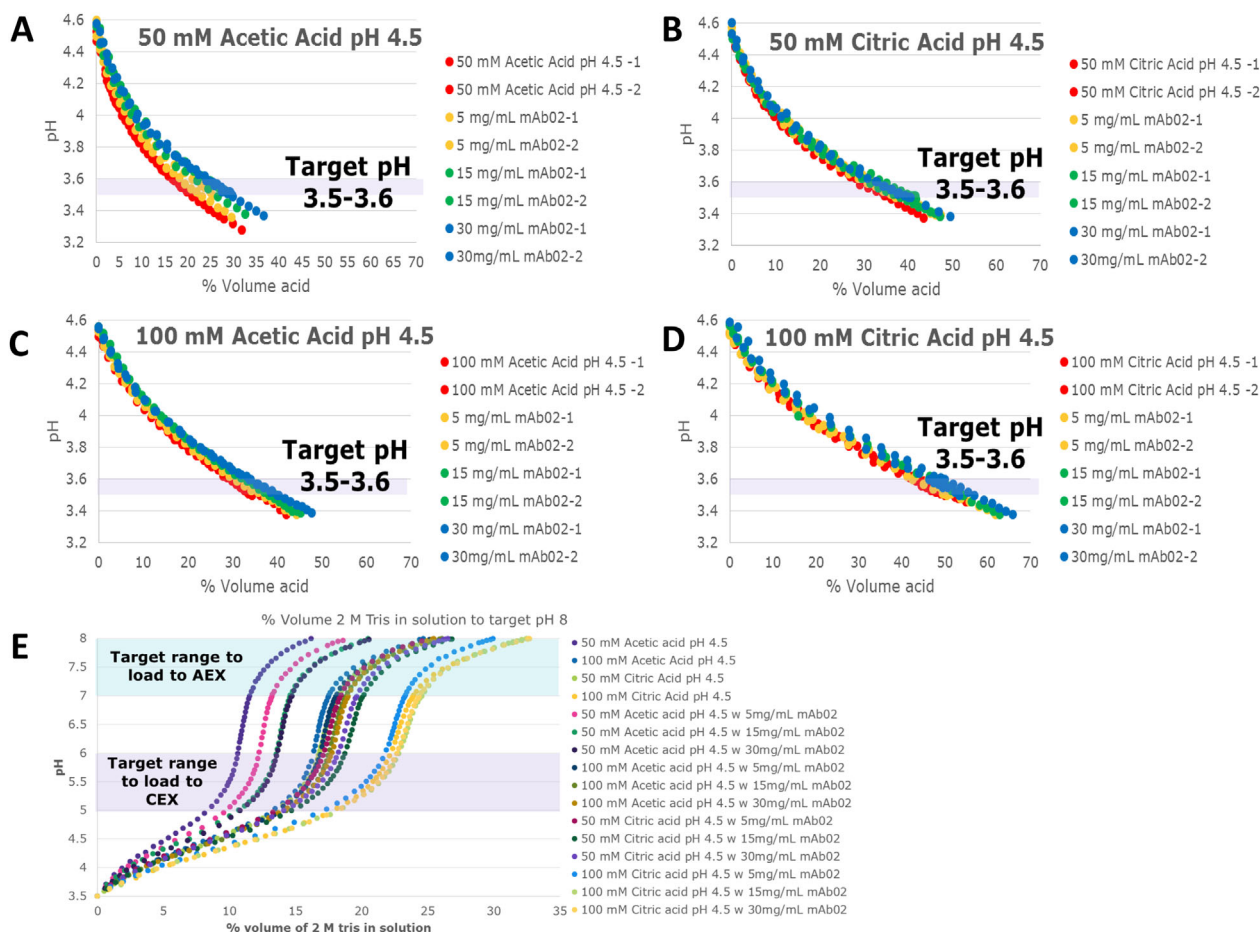


Figure 3. Acid and base titration data for several buffer systems and mAb concentrations. Titrations were performed with 1 M acetic acid. Duplicate experiments are shown. A) 50 mM acetic acid, pH 4.5, (B) 50 mM citric acid, pH 4.5, (C) 100 mM acetic acid, pH 4.5, (D) 100 mM citric acid, pH 4.5, (E) 2 M Tris base titration to reach pH 8 target.

Initial studies were performed using bacteriophage Phi6 as a model virus, due to ease-of-use and faster time to result. To characterize the Phi6 pH inactivation profile, batch-mode screening studies were performed at a pH range of 3.6–5.5 while the Control was maintained at pH 6.2. As described above, pH was adjusted first and then virus was spiked into the Test Material and Control and vortexed to ensure homogeneous mixing. After 1 and

30 min, 1 mL samples were removed and neutralized to pH > 6.2 with 2 M Tris base, immediately diluted in Phi6 assay buffer and assayed. The results, shown in **Figure 5A**, demonstrate that Phi6 is rapidly and completely inactivated at pH 3.5 within 1 min, but at higher pH levels, inactivation was slower and less complete. At pH 5.5, the Phi6 titer is the same as the Control titer indicating no viral inactivation. Based upon these batch screening studies, Phi6 in-line inactivation studies were performed at a pH target of 3.5–3.6.

Consistent with the batch-mode study, complete Phi6 inactivation was observed with the in-line system within 1 min at pH 3.6, shown in **Figure 5B**.

To better quantify the in-line system's viral inactivation capability compared to batch-mode, a study was carried out at an "intermediate" or worst-case pH that would provide slower inactivation kinetics and enable measurement of differences between the two systems based on changes in viral titer. Based upon the results of Phi6 inactivation kinetics at various pH in batch screening studies, (Figure 5A) an intermediate pH of 4.6 was selected for further testing. The Phi6 inactivation results show good agreement between the batch and in-line systems, with the titers at most of the timepoints differing by less than 1 log, as shown in **Figure 5C**, while slightly larger deviations are observed at the longer time scales.

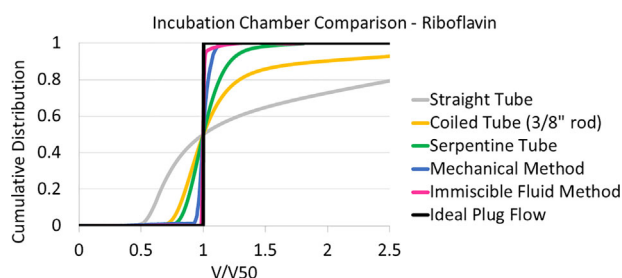


Figure 4. Cumulative distribution plots for different incubation chamber designs. Results are for pulse injections of riboflavin detected by UV absorbance at incubation chamber outlet. Data are normalized to V50 values (volume at 50%). Ideal plug flow is shown for comparison.

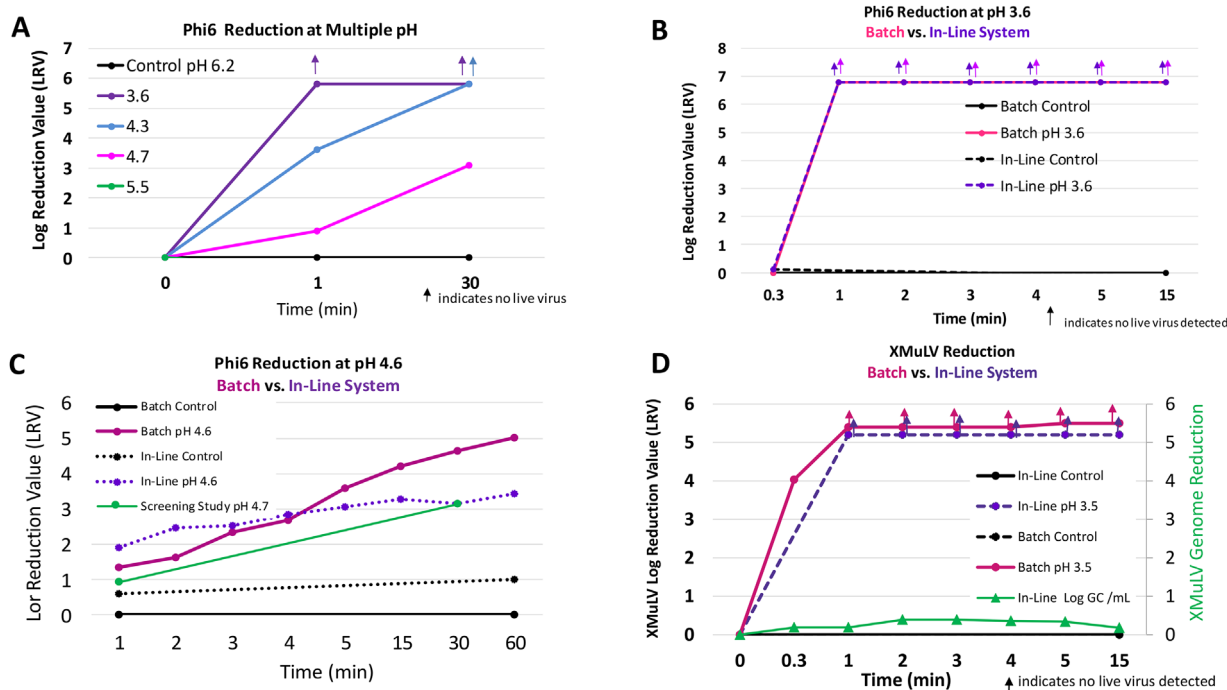


Figure 5. Batch and in-line virus inactivation results for two viruses. Virus LRVs are shown as a function of time. Arrows indicate that no live virus was detected. A) Batch Phi6 inactivation at multiple pH values, (B) Batch and in-line Phi6 inactivation at pH 3.6, (C) Batch and in-line Phi6 inactivation at pH 4.6–4.7, (D) Batch and in-line XMuvLV inactivation at pH 3.5; qPCR results are shown as the reduction of genome copies mL^{-1} .

A similar comparison study of batch and in-line viral inactivation was performed using XMuvLV at pH 3.5, as shown in Figure 5D. Results show XMuvLV inactivation is rapid and complete within 1 min with an LRV > 5 for both batch and in-line systems. Due to constraints posed by the system, 1 min is the shortest possible sampling point for the in-line system, while 0.3 min is the first sampling point for the batch format, the time required to spike virus, vortex, remove a sample and neutralize. qPCR analysis confirmed no change in viral particle concentration following processing through the in-line system.

4. Discussion

Continuous processes necessitate a re-evaluation of processing operations to enable the changes needed to streamline production, and achieve increased efficiency while continuing to meet both the purification and safety needs of the production process. Improved understanding of the important process parameters can inform operational changes.

Virus inactivation was evaluated under a wide range of conditions to assess the impact of different buffer conditions on the amount of acid and base required to achieve target values. These findings were used as initial setpoints for in-line pH adjustments as the complexity of a Protein A elution pool may require modeling to accurately describe and predict the behavior that will result as a function of different buffer types and concentrations, temperature, as well as mAb concentrations.

Numerous studies with multiple viruses were generated to demonstrate very rapid low pH inactivation. These included a range of pH levels, buffer compositions, mAb concentrations,

and temperatures. These data indicate that complete viral inactivation is typically achieved within minutes, which is a much shorter time scale than traditional viral inactivation process of 30–60 min. With an efficient incubation chamber that can achieve rapid viral inactivation, including an appropriate safety factor, the in-line inactivation operation could offer considerable process flexibility and reduced facility footprint compared to the traditional batch process.

A variety of incubation chamber designs were developed and tested. The coiled tub and serpentine tube designs use coils to induce secondary flows which pull the fluid away from the walls and enhance radial mixing. Although these designs offered improved performance compared to a straight tube, the best performing designs were based on the idea of creating packets of fluid. One creates packets using a mechanical apparatus, analogous to the method of a peristaltic pump (see the Materials and Methods section for more detail) and the other method uses an immiscible fluid to separate packets of fluid. In the case of an immiscible fluid the packets are separated by air or another fluid to discretize the fluid into smaller packets where the immiscible fluid is later removed after the fluid exits the chamber. Both designs result in narrow residence time distributions which approach the ideal plug flow performance due to the minimal axial mixing that can only occur within the packet as it traverses the incubation chamber.

Straight tube in-line low pH inactivation of two viruses was performed and the results are comparable to the batch system. Further analysis at an intermediate pH level enabled a more quantitative comparison and demonstrated equivalence between lab-scale batch and in-line systems.

The feasibility of continuous in-line virus inactivation has been demonstrated, with future work intended to improve

incubation chamber design to optimize system fluidics pathway and assess any potential impact on product quality. These results are consistent with data available in literature.^[9] A comprehensive risk analysis and manufacturability assessment, particularly on the incubation chamber design options and potential validation strategies, is crucial for guiding decisions for future system design and scale-up options.

5. Conclusion

In this investigation, rapid viral inactivation was demonstrated with multiple viruses under a variety of conditions. The impact of various solution conditions on pH was also assessed through acid and base titrations. Efficient incubation chamber designs were developed which offered residence time distributions that approached the ideal case of plug flow. A lab-scale in-line virus inactivation system was designed, built, and characterized. Equivalency between batch and in-line systems was demonstrated. Overall, this work offers data that indicates this approach can be successfully designed and implemented for in-line virus inactivation in integrated and continuous mAb purification processes.

Abbreviations

hIgG, human Immunoglobulin G; LRV, log reduction value; mAb, monoclonal antibody; Phi6, Phi6 bacteriophage; PRV, pseudorabies virus; TCID₅₀, 50% Tissue Culture Infectious Dose; XMuLV, xenotropic murine leukemia virus.

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Conflict of Interest

The authors declare no financial or commercial conflict of interest.

Keywords

continuous, inactivation, incubation chamber, kinetics, virus

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